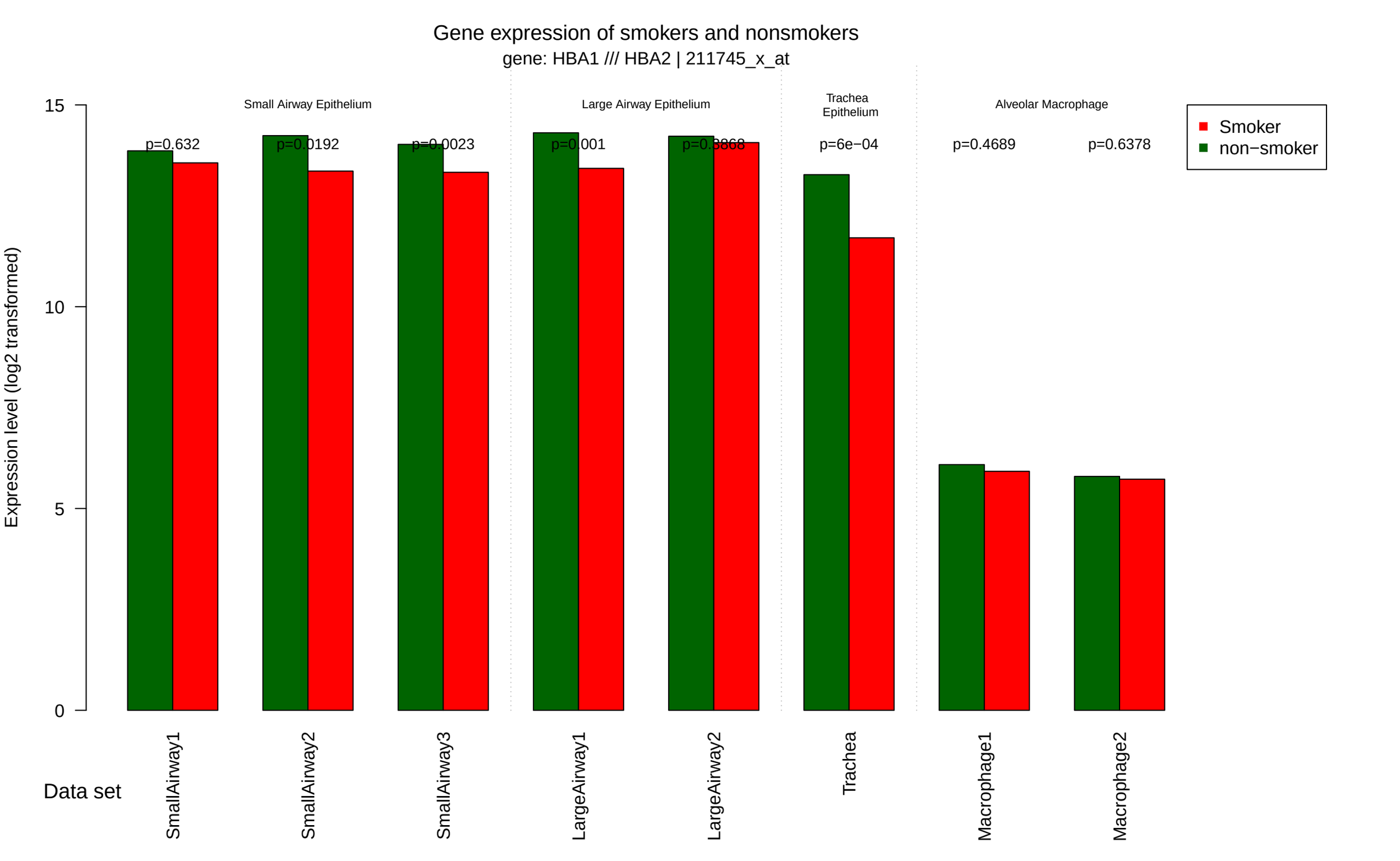
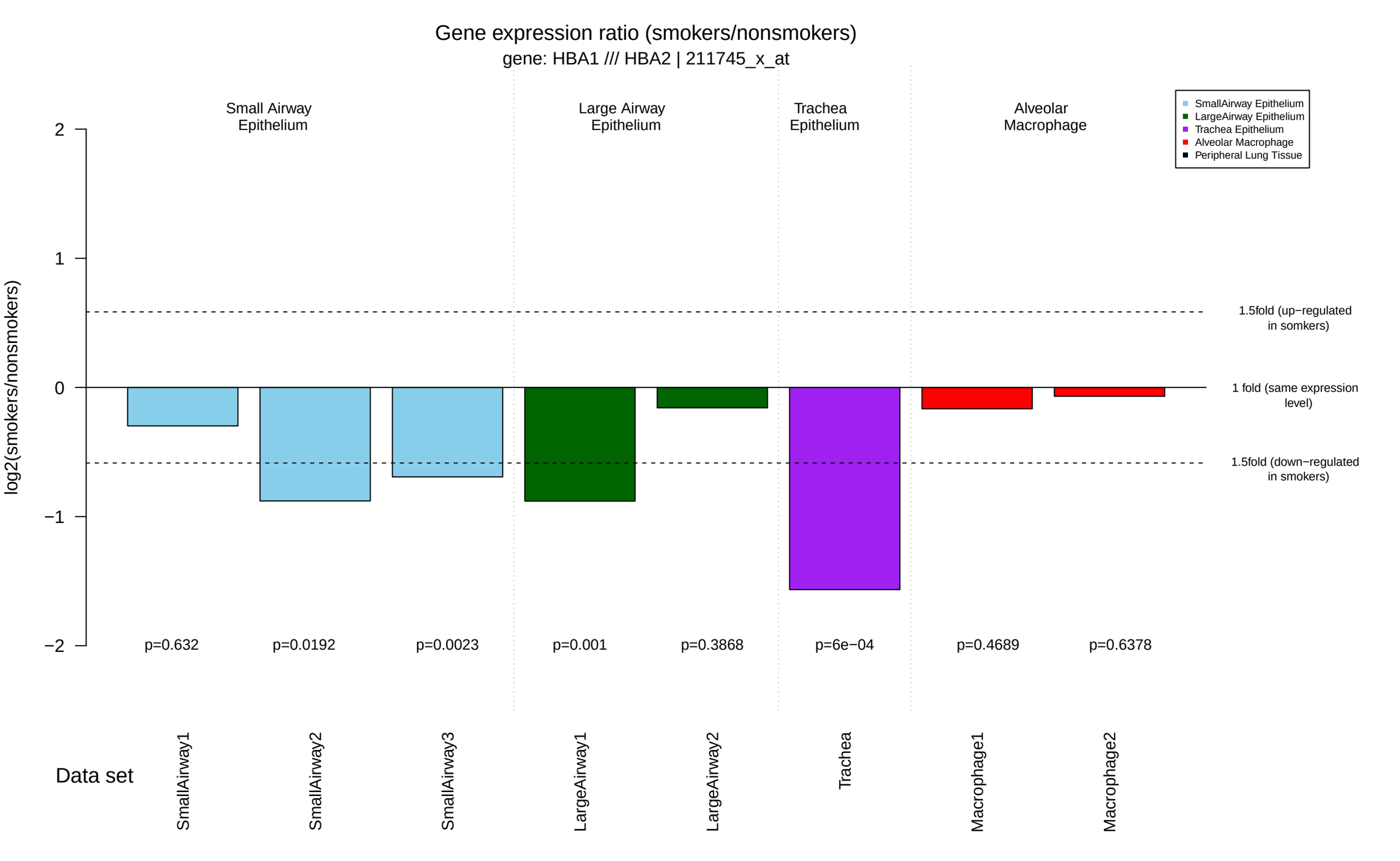
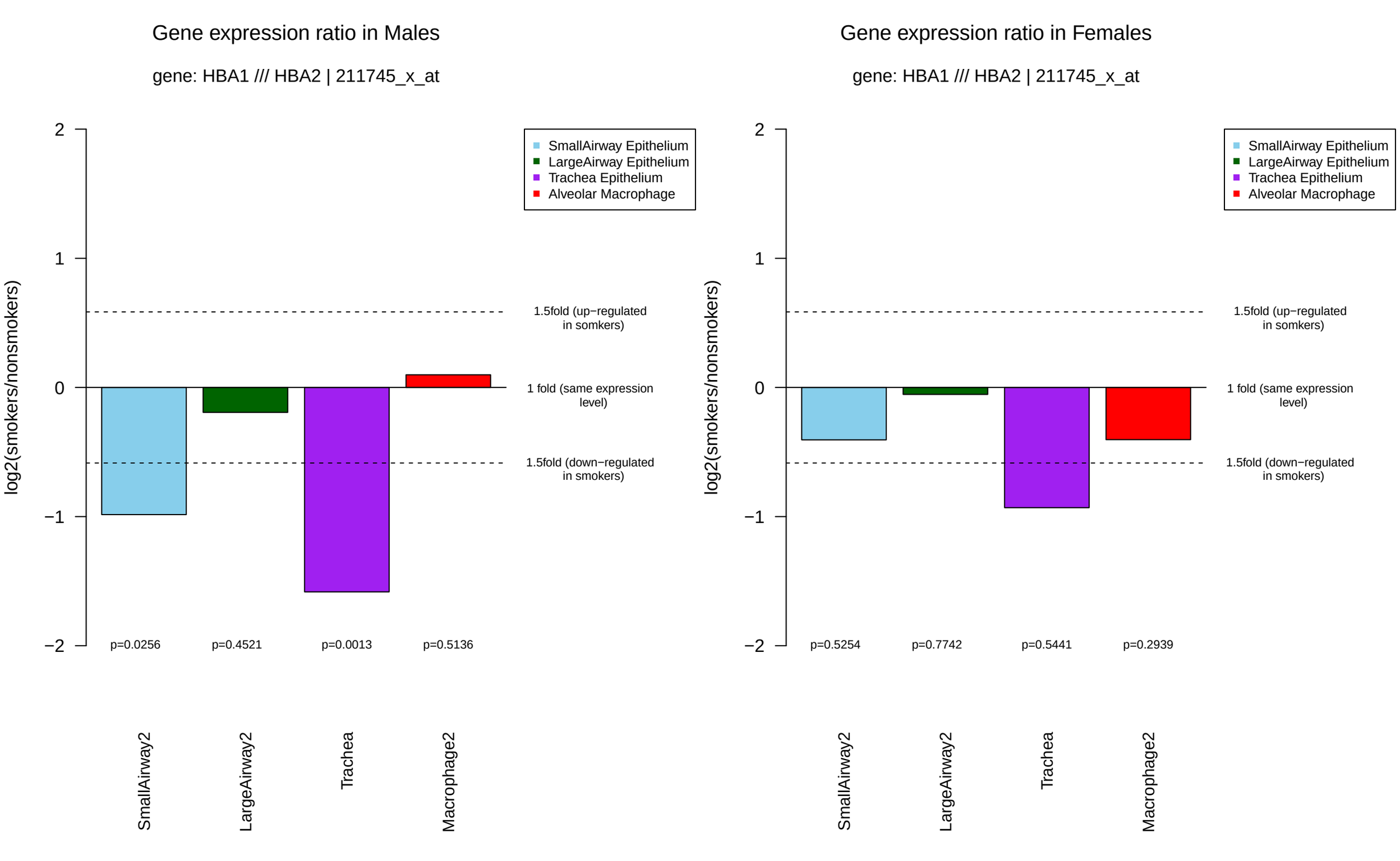
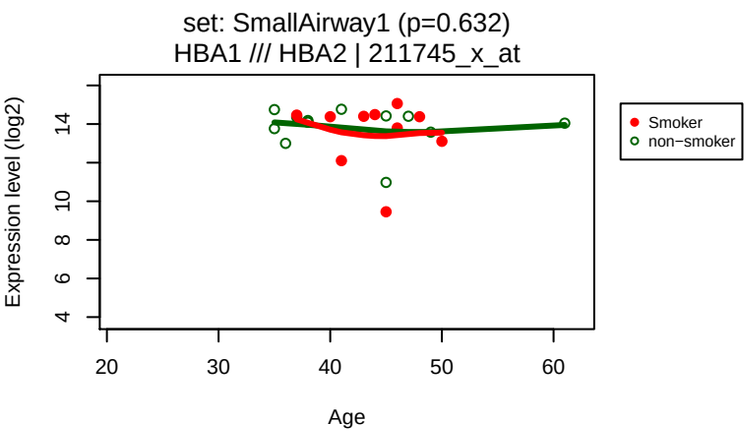
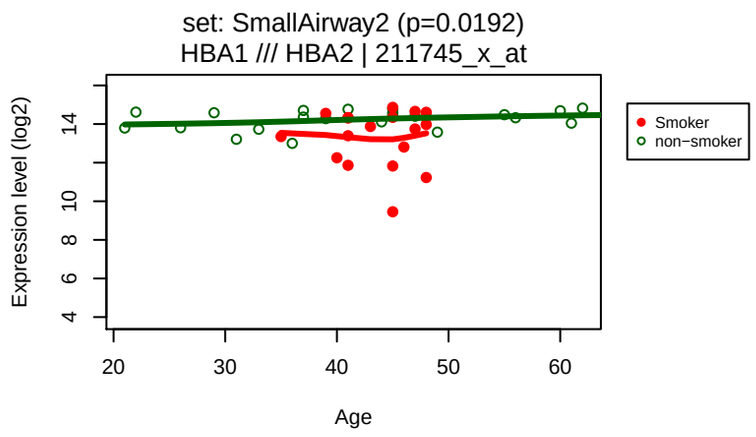
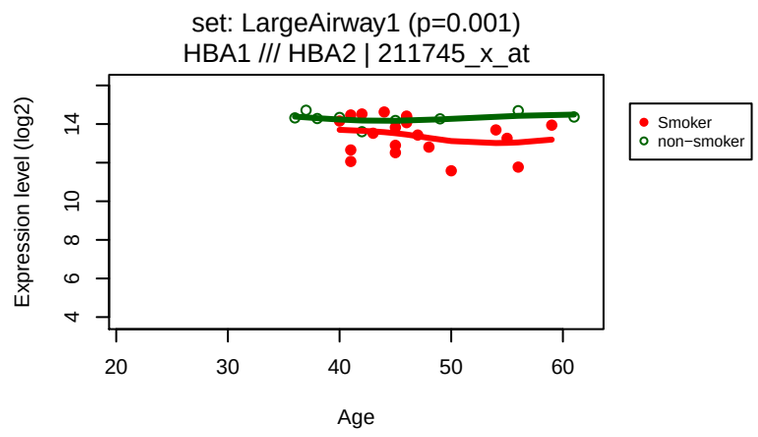

Gene expression of smokers and nonsmokers
gene: HBA1 /// HBA2 | 211745_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.632
p=0.0192
p=0.0023
p=0.001
p=0.3868
p=6e−04
p=0.4689
p=0.6378
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

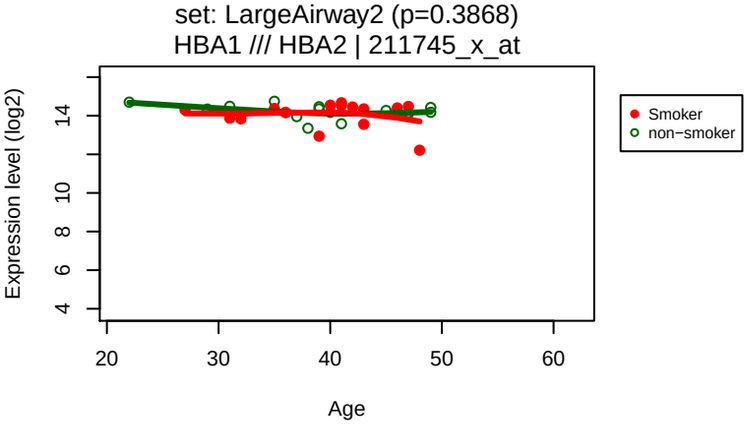

Gene expression ratio (smokers/nonsmokers)
gene: HBA1 /// HBA2 | 211745_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.632
p=0.0192
p=0.0023
p=0.001
p=0.3868
p=6e−04
p=0.4689
p=0.6378
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

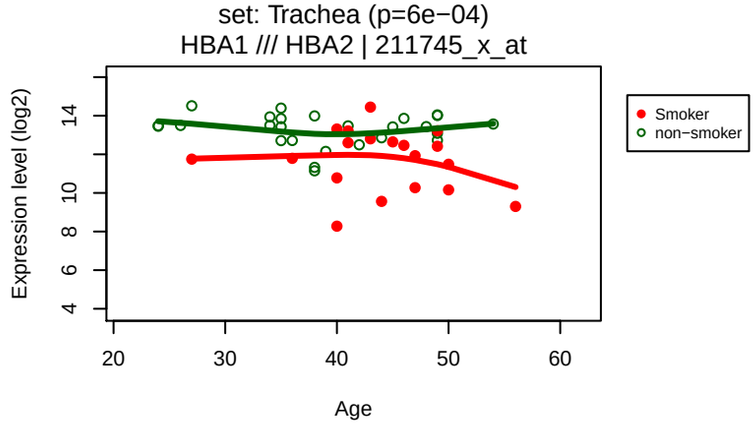

Gene expression ratio in Males
gene: HBA1 /// HBA2 | 211745_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0256
p=0.4521
p=0.0013
p=0.5136
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HBA1 /// HBA2 | 211745_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5254
p=0.7742
p=0.5441
p=0.2939
SmallAirway2
LargeAirway2
Trachea
Macrophage2

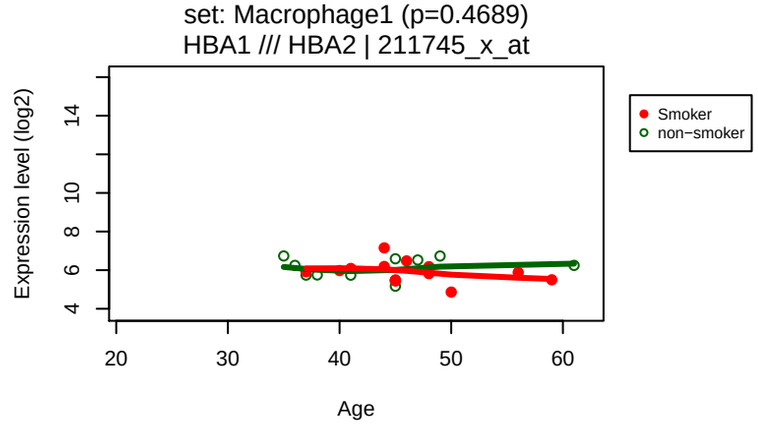

set: SmallAirway1 (p=0.632)
HBA1 /// HBA2 | 211745_x_at
Expression level (log2)
Age
Smoker
non−smoker

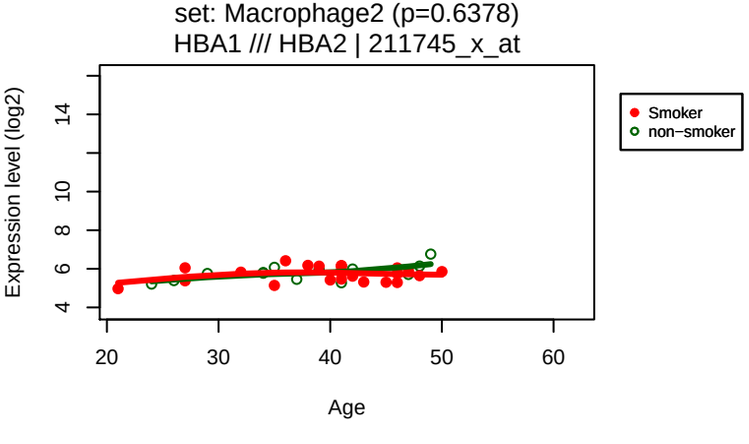

set: SmallAirway2 (p=0.0192)
HBA1 /// HBA2 | 211745_x_at
Expression level (log2)
Age
Smoker
non−smoker

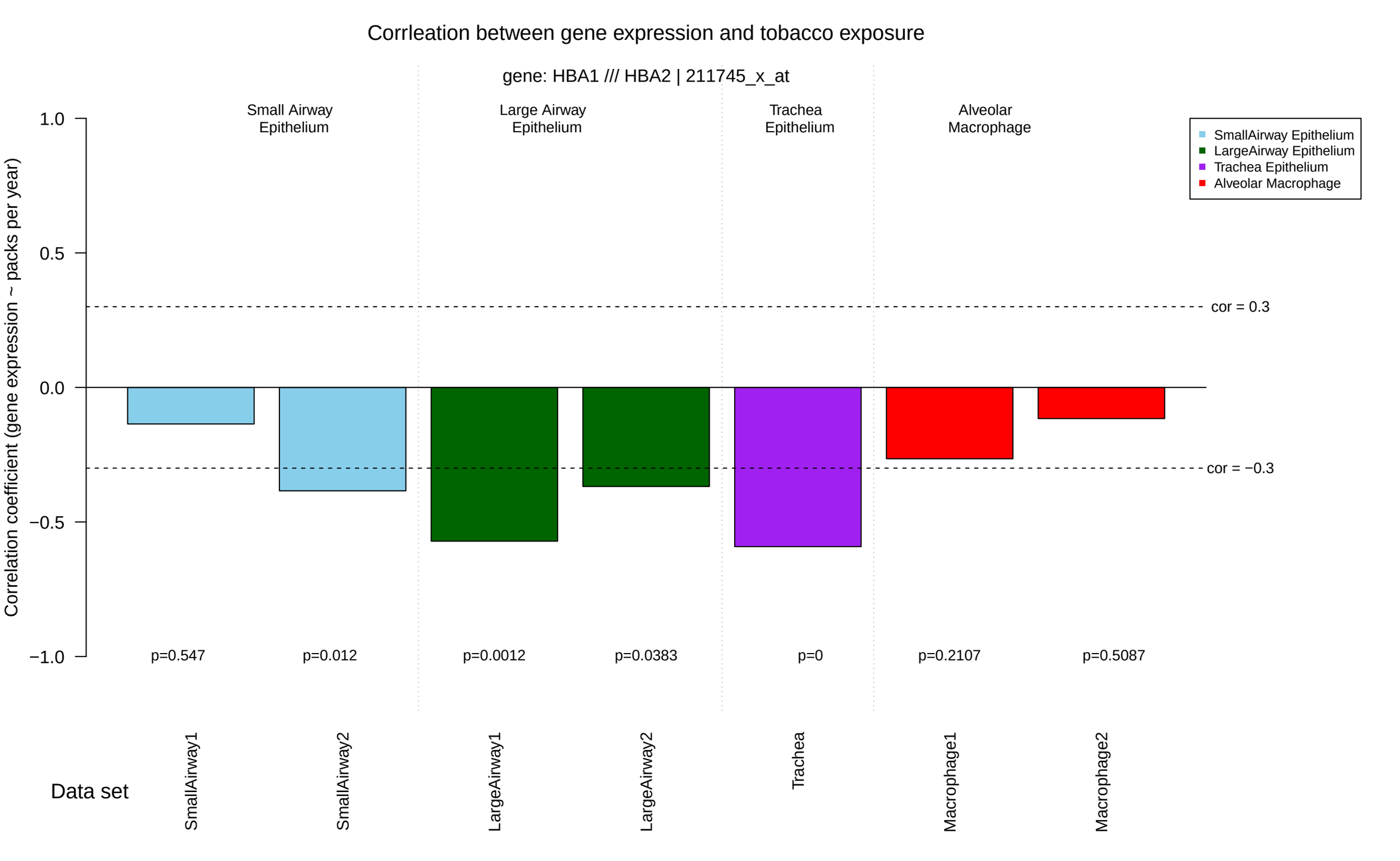

set: LargeAirway1 (p=0.001)
HBA1 /// HBA2 | 211745_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3868)
HBA1 /// HBA2 | 211745_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=6e−04)
HBA1 /// HBA2 | 211745_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4689)
HBA1 /// HBA2 | 211745_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6378)
HBA1 /// HBA2 | 211745_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HBA1 /// HBA2 | 211745_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.547
p=0.012
p=0.0012
p=0.0383
p=0
p=0.2107
p=0.5087
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2